

Two Blind Spots in Backtest Validation: Evidence from the U.S. Dollar Index

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Abstract

Backtest validation is built almost entirely around one failure, that too many strategies were tried. Using the U.S. Dollar Index and its constituent currency futures over 2010 to 2025, we document two failures this defence cannot detect. First, a widely used free data source records session opening prices with a stale print defect that manufactures an overnight anomaly of Sharpe -1.36 ($p < 0.0001$) absent from exchange futures on the same instrument over the same sessions. The standard diagnostics do not merely miss the artefact, they certify it. The probability of backtest overfitting is 0.02 for the false result against 0.77 for the correct data and before costs the artefact survives stepwise familywise error control, because these tools discipline the selection of strategies and never inspect the inputs. An exact binomial screen on the location of the open within the daily range detects the defect at $p = 3.7 \times 10^{-13}$ with no external reference series. Second, a regime conditioned currency strategy reporting Sharpe 0.72 (Newey–West $t = 2.78$) reverses once its full 216 configuration search is declared instead of the six reported. A correctly specified search of 1,312 configurations drawn from published specifications then finds none exceeding the Sharpe ratio the same search attains on block bootstrapped returns. Reporting that noise ceiling would have exposed both failures at once.

Keywords: backtest overfitting; data quality; deflated Sharpe ratio; foreign exchange; multiple testing; reproducibility; trading strategy evaluation

JEL Classification: C12; C15; C52; F31; G14

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1. Introduction

The modern defence of a quantitative backtest is a defence against one specific failure. The analyst enumerates the strategies that were tried, then applies a correction for the number of trials: White’s Reality Check (White, 2000), Hansen’s test of superior predictive ability (Hansen, 2005), the stepwise procedure of Romano and Wolf (2005), the deflated Sharpe ratio of Bailey and Lopez de Prado (2014), the probability of backtest overfitting of Bailey et al. (2017), or the haircut Sharpe ratio of Harvey and Liu (2015). This machinery is well developed and widely adopted.

It is also narrow. Every one of these tools takes the input data as given and asks a question about the strategy set. Two failures therefore sit outside its field of view. The input series may be wrong, in which case the correction is applied faithfully to a number that never existed. Or the declared strategy set may be smaller than the set actually explored, in which case the correction is applied to the wrong denominator.

This paper documents one of each, on the same instrument, with one codebase and then shows what a correctly specified search on the same market produces. Our contribution is not that vendor data contains errors. Ince and Porter (2006) established that for equity returns two decades ago and their paper remains the standard reference. Our contribution is what happens when that literature is placed next to the overfitting diagnostics that postdate it, in particular the deflated Sharpe ratio and the probability of backtest overfitting. The answer is that those diagnostics invert. On the corrupted series the probability of backtest overfitting is 0.016 to 0.028, which reads as an exceptionally robust result. On the correct series for the same instrument it is 0.765 to 0.779, which reads as pure noise fitting. Both readings are internally

correct. A stationary data defect keeps winning out of sample for exactly the same reason a real anomaly does, so no overfitting diagnostic can separate them.

Section 2 describes the data. Section 3 isolates the defect and shows what the diagnostics say about it, then gives a screen that detects it. Section 4 documents the second failure, an undeclared search space. Section 5 runs a correctly specified search and introduces the noise ceiling. Section 6 asks whether anything in this market is profitable once the analysis is driven by structure rather than by enumeration. Section 7 states the reporting protocol we recommend and Section 8 the limitations.

2. Data

Table 1 lists the four sources. The two U.S. Dollar Index series are the centre of the paper. One is the ICE spot index as redistributed by a free provider, the series on which the original result was found. The other is the ICE Dollar Index futures contract obtained from a commercial vendor sourced from the exchange. They describe the same underlying index over the same dates.

Table 1. Data sources. The two dollar index series in the first two rows describe the same underlying index over the same dates and are the comparison on which Section 3 rests.

Series	Source	Period	Sessions
Dollar index, free provider	Free redistribution of the ICE spot index	2010-12 to 2025-12	3,932
Dollar index futures	Commercial vendor, exchange sourced (DX)	2010-12 to 2025-12	3,889
Currency futures	CME contracts E6, B6, D6, J6, S6	2010-12 to 2025-12	3,800
Spot rates and interest rates	Federal Reserve H.10 and OECD via FRED	1999-01 to 2026-08	6,929

The five currency futures are the CME contracts for the euro, sterling, Canadian dollar, yen and Swiss franc. No liquid Swedish krona contract exists, so the tradeable universe is five of the six index constituents. All five carry median daily volume above 25,000 contracts, contain no violation of the open, high, low, close ordering and are correctly dated: their return correlation with the Federal Reserve noon fix peaks at lag zero.

Nearby continuous futures jump when the front contract changes. That jump is the spread between two contracts and not a return any holder earns, so all returns are computed within contract and the 61 roll sessions in each series are assigned a return of zero. This also means that contract changes cannot contribute to the overnight leg examined in Section 3.

Every t statistic reported below is corrected for heteroskedasticity and autocorrelation following Newey and West (1987), using a Bartlett kernel with the standard automatic bandwidth $\lfloor 4(T/100)^{2/9} \rfloor$, which gives eight lags at this sample length. The conclusions are insensitive to that choice: the overnight leg of the free series returns $p < 0.0001$ at eight, nine and twelve lags and the exchange series returns 0.328, 0.328 and 0.331.

3. A Data Defect That the Diagnostics Certify

3.1. The result and the first attempt to explain it

Decomposing the daily return into an overnight leg from the previous close to the open and a daytime leg from the open to the close produces, on the free series, an overnight leg with a Sharpe ratio of -1.36 and $p < 0.0001$. This is a large effect. On the exchange futures for the same instrument the overnight leg has a Sharpe ratio of $+0.25$ with $p = 0.328$. Both figures are computed on the 3,793 overnight returns available in the 3,794 sessions for which both series report a complete bar, so the two are directly comparable.

A naive way to locate the source is to substitute one vendor's opening prices into the other vendor's series. That test is invalid here. The two series carry a persistent level offset of 0.041 index points, which is 6.7 per cent of a typical daily range, so a price level substitution injects the offset into every return. Performing it produced a terminal value of 1178 for one strategy, which is not a finding but an artefact of the offset.

3.2. Isolating the open

The valid version transfers the close to open *return* rather than the price, so that each row uses one vendor's price level throughout and the result is free of the offset. Table 2 reports it.

Table 2. Scale free decomposition. Each row uses a single vendor's price level throughout and transfers only the close to open return, so the level offset between vendors cannot contribute. The daytime leg is the residual, which makes the 24 hour return identical across rows by construction. Terminal values start from 100 and the sample is the 3,793 overnight returns common to both series.

Price level, opening return	Overnight leg			Daytime leg		
	Final	Sharpe	p	Final	Sharpe	p
Free, free open	61.68	−1.358	0.0000	197.35	0.700	0.0086
Free, exchange open	108.08	+0.252	0.3280	112.62	0.151	0.5534
Exchange, exchange open	108.08	+0.252	0.3280	112.22	0.146	0.5668
Exchange, free open	61.68	−1.358	0.0000	196.65	0.678	0.0103

The overnight leg follows whichever vendor's open is used and is unchanged by whose closes are used. This is as close to a controlled experiment as observational data allows. The opening prices are the entire source of the effect.

Table 3 shows what is wrong with them. With the level offset removed, the free series' open lies outside the exchange's true high to low range on 7.30 per cent of sessions. On the 204 sessions where the free open equals its own daily low it sits below the exchange low, at −0.073 of the exchange range and falls outside that range on 56.9 per cent of those sessions against 3.5 per cent on ordinary sessions. A price outside the session range is not a price at which the instrument traded. The free series also records an open exactly equal to the previous close on 6.84 per cent of sessions against 1.34 per cent for the exchange series, which is the signature of a stale print.

Table 3. Location of the recorded open within the exchange session range, with the level offset removed. A value of 0 is the exchange low and 1 the exchange high. The sessions on which the free open equals its own extreme are the sessions that carry the anomaly.

Session group	n	Open location	Outside range
Free open at its own low	204	−0.073	56.9%
Free open at its own high	82	+0.994	45.1%
All other sessions	3,508	+0.494	3.5%
Exchange open, all sessions	3,889	+0.498	0.00%

3.3. What the diagnostics say

The 24 strategy grid of the original study was rerun on both series. On exchange data no strategy beats buy and hold at any cost level, the analysis of variance across the 24 gives $F(23, 93312) = 0.306$ with $p = 0.9994$ and the largest Sharpe ratio in the grid, +0.269, is below the +0.316 that the null expects the maximum of 24 trials to produce. The walk forward over 13 out of sample years ends at 97.69 against buy and hold at 122.64. There is nothing in the exchange data.

Table 4 is the central result of this section. Applied to the corrupted series at zero cost, Hansen's test rejects at $p = 0.013$, three configurations survive the stepwise procedure of Romano and Wolf (2005) and the probability of backtest overfitting is 0.016. Applied to the correct series the same diagnostics report 1.000, zero survivors and 0.779. The diagnostics rank the false result far above the true one.

Table 4. Multiple testing and overfitting diagnostics applied to the same 24 strategy grid on the two dollar index series, at zero transaction cost. RW is the number of configurations surviving the stepwise familywise error procedure of Romano and Wolf (2005). PBO is the probability of backtest overfitting from the combinatorially symmetric cross validation of Bailey et al. (2017), reported at 12 and 16 blocks. Lower PBO indicates a result that persists out of sample.

Series	White RC p	Hansen SPA p	RW survivors	PBO
Free provider (the artefact)	0.1032	0.0132	3 of 24	0.016 to 0.028
Exchange futures (correct)	0.9924	1.0000	0 of 24	0.765 to 0.779

Both readings are correct on their own terms. The probability of backtest overfitting asks whether a strategy selected in sample continues to win out of sample. A persistent data defect continues to win, because it is present in both halves of the sample. To every overfitting diagnostic a stationary artefact and a real anomaly are the same object. This is a structural limitation of the entire family of tools and not a defect in any one of them.

3.4. A screen that requires no reference data

The comparison above required a second, trustworthy series. Most analysts do not have one. We therefore propose a test that uses only the series being examined.

Under the null that a recorded open is an unbiased draw from the prices visited during the session, the open is equally likely to coincide with the session low as with the session high. Counting the two events gives an exact binomial test with no free parameters:

$$n_{\text{low}} \sim \text{Binomial}\left(n_{\text{low}} + n_{\text{high}}, \frac{1}{2}\right). \quad (1)$$

Table 5 applies Equation (1) to eight series. It clears the exchange series at $p = 0.788$ and flags the corrupted one at $p = 3.7 \times 10^{-13}$.

Table 5. The open location binomial screen of Equation (1) applied to eight series. Mean position is the average location of the open within the session range, where 0 is the low and 1 the high. The screen clears the series known to be sound and rejects the series known to be corrupt, using only that series' own open, high and low.

Series	At low	At high	Binomial p	Mean position
Dollar index futures, exchange	64	60	0.788	0.499
Dollar index, free provider	204	82	3.68×10^{-13}	0.479
EURUSD, free provider	90	56	0.0061	0.501
GBPUSD, free provider	69	58	0.375	0.504
JPYUSD, free provider	125	103	0.164	0.489
CADUSD, free provider	66	52	0.231	0.506
CHFUSD, free provider	62	44	0.098	0.505
SEKUSD, free provider	44	45	1.000	0.508

Two honest qualifications. The screen also flags EURUSD at $p = 0.0061$, which barely survives a Bonferroni correction across the eight tests at a threshold of 0.00625. We independently established that these spot series are date stamped one trading day late, which is consistent with a true positive, but we cannot prove it is not a multiplicity artefact. And the screen is validated here against exactly one instrument with a known good reference. It is a promising diagnostic and not an established one.

We note also that the mean position of the open within the range, at 0.479 for the corrupted series against 0.499 for the exchange series, is a weak discriminator. The count of extremes carries the information, not the average location.

4. An Undeclared Search Space

The second failure needs no bad data. A regime conditioned cross sectional currency strategy, in which six currencies are ranked each quarter and the two weakest are held with a sign set by the 126 day momentum of the dollar index, reports a Sharpe ratio of 0.72 with a Newey–West t statistic of 2.78 and $p = 0.0055$ on Federal Reserve spot data.

Three defects in the committed implementation had to be corrected first. The pandas call `to_period('2Q')` silently returns quarterly periods, so the reported semiannual row was a byte identical duplicate of the quarterly row. The grid loop was wrapped in a bare exception handler, which concealed that the annual frequency alias had been renamed between pandas versions, so every annual configuration was silently discarded. The construction `np.where(mom < 0, 1, -1)` maps a missing value to -1 , so the first 126 sessions, in which no momentum can yet be computed, were traded as though the dollar were rising; correcting this alone moves the headline Sharpe ratio from 0.54 to 0.46. And the constituent price files are date stamped one trading day late relative to the index file.

Table 6 reports what happens when the search space is declared in full. The original account reported six configurations. Frequency, cross sectional slice, cardinality, the regime filter and the momentum lookback are all free parameters and enumerating them gives 144 configurations on the tradeable five currency panel and 216 on the six currency panels.

Table 6. Multiple testing over the declared search space. The strategy is a self financing long and short book, so the correct null is zero excess return; the committed code tested against a long dollar position, which is shown for comparison. The number after each universe is the count of currencies in it. No configuration survives the stepwise procedure in any universe once the full search is declared.

Universe	Configs	Sharpe	Null: zero		Null: long dollar		DSR
			RC p	SPA p	RC p	SPA p	
Exchange futures, 5	144	0.458	0.2333	0.3833	0.6730	0.7647	0.227
Futures plus krona, 6	216	0.498	0.2203	0.2830	0.4373	0.5933	0.217
Federal Reserve spot, 6	216	0.722	0.0963	0.1570	0.3173	0.4727	0.613

The deflated Sharpe ratio is a confidence level and must exceed 0.95. It reaches 0.613 at best. The expected maximum Sharpe ratio under the null across 144 configurations is $+0.654$, above the $+0.458$ actually observed on tradeable instruments. No universe now clears a conventional threshold on any test.

The configuration is also a spike rather than a plateau. Holding everything else at the reported values on the tradeable five currency panel and moving one parameter at a time, the regime lookback gives -0.319 with no overlay, $+0.136$ at 63 days, $+0.458$ at 126 days and $+0.259$ at 252 days. The rebalance frequency gives -0.234 daily, -0.081 weekly, $+0.047$ monthly, $+0.458$ quarterly, $+0.112$ semiannually and $+0.456$ annually. Only two of six frequencies are materially positive and the neighbours of the reported lookback lose most of the effect, which describes a noisy surface rather than a mechanism.

Finally, an ensemble that holds every configuration in a family at equal weight selects nothing and therefore requires no correction. Averaging the net returns of the 36 regime signed configurations in the bottom slice, which charges each member its own full turnover and so is conservative, gives a Sharpe ratio of 0.148 with $p = 0.57$. Widening the ensemble to all 108 regime signed configurations gives -0.160 with $p = 0.53$.

5. A Correctly Specified Search

5.1. Validating the harness before believing it

Before running any search we validated the evaluation code itself, because a paper about invalid backtests should demonstrate that its own backtest is valid. Table 7 lists the ten checks. The most important is the first. Weights formed from the same day's realised return, applied without the one period shift, must produce an absurd Sharpe ratio; if they do not, the shift is broken and nothing downstream means anything.

Table 7. Validation of the evaluation harness. All ten checks pass; the eight rows below group the four turnover and statistic checks into two lines. The lookahead injection is the critical one, because the harness must be demonstrably capable of detecting a strategy that cheats.

Check	Result
Perfect foresight weights applied without the shift	Sharpe +24.34
The same weights applied through the harness	Sharpe −0.11
One extra lag applied to 14 real families	none collapses
Full search on simulated zero drift returns, RC p	0.20, 0.07, 0.69, 0.38
Observed best of N on noise against theory	0.823 against 0.846
Sharpe, Newey–West t , PSR, cost arithmetic	reproduced to machine precision
Buy and hold turnover; daily sign flipper turnover	0.00; exactly 2.0 per session
Winning configuration rebuilt from first principles	maximum difference 0.0e+00

5.2. The search and the noise ceiling

We evaluated 1,312 configurations built from published specifications: time series momentum (Moskowitz et al., 2012), cross sectional currency momentum (Menkhoff et al., 2012), carry (Lustig et al., 2011; Kojien et al., 2018), value (Asness et al., 2013), moving average rules (Brock et al., 1992), channel breakouts, open interest growth (Hong and Yogo, 2012) and volatility scaling (Moreira and Muir, 2017), each crossed with four rebalance frequencies, the regime overlay at three lookbacks and with or without volatility targeting. Costs are charged at one basis point per unit of turnover.

The natural benchmark is not zero. It is the Sharpe ratio this same search attains on returns that contain no predictable structure. We construct that benchmark by resampling the actual returns in blocks using the stationary bootstrap of Politis and Romano (1994), which preserves the marginal distributions, the fat tails and the cross sectional correlation while destroying the time series structure a signal could exploit, then running the entire search on the resampled panel. We call the resulting maximum the *noise ceiling*. It is not a new statistic. It is a Monte Carlo reality check expressed in the units a practitioner reads.

Table 8. The correctly specified search against its noise ceiling. The ceiling is the best Sharpe ratio the identical 1,312 configuration search attains on block bootstrapped returns, averaged over six resamples. No configuration on the real data exceeds it and every correction agrees.

Quantity	Value
Best configuration on the real data, net of costs	+0.653
Noise ceiling, Gaussian null	+0.752
Noise ceiling, block bootstrap of the real returns	+0.712
Expected best of 1,312 under the null	+0.865
Configurations exceeding the ceiling	0 of 1,312
White Reality Check p	0.659
Hansen SPA p	0.773
Romano–Wolf survivors	0
Deflated Sharpe ratio (needs > 0.95)	0.203
Harvey–Liu adjusted p , Bonferroni and BHY	1.000, 1.000
Probability of backtest overfitting	0.556

Table 8 gives the result. The best configuration found in fifteen years of exchange data is below what the same search finds in random data. The holdout is consistent: selecting on 2011 to 2018 and evaluating once on 2019 to 2025, the in sample winner ranks 163rd of 1,312 out of sample and the top ten in sample configurations average a Sharpe ratio of +0.019.

Family ensembles, which select nothing, are reported in Table 9. Not one is significant in either direction.

Table 9. Family ensembles, equal weight within each family, costed in weight space so that trades offsetting across members are not charged. Selecting nothing means no correction is required, so these are the fair test of whether a family of ideas has value.

Family	Sharpe	p	Family	Sharpe	p
Cross sectional momentum	+0.218	0.37	Time series momentum	−0.214	0.37
Value	+0.204	0.43	Volume	−0.232	0.37
Carry	+0.135	0.59	Breakout	−0.289	0.23
Reversal	+0.133	0.58	Moving average	−0.341	0.16
Open interest	+0.094	0.72	Cross sectional reversal	−0.360	0.14
All 1,312 configurations				−0.176	0.48

6. Is Anything Here Profitable?

An exhaustive search that finds nothing invites the objection that the search was the wrong instrument. We therefore also proceeded analytically. The structure of the data is unambiguous. The first order autocorrelation of returns is between -0.004 and $+0.033$, variance ratios lie between 0.83 and 1.02 and an autoregression of returns on their own lag gives R^2 between 0.0000 and 0.0011 . The same autoregression on log realised volatility gives R^2 between 0.966 and 0.972 . Volatility is roughly three orders of magnitude more forecastable than direction.

If direction cannot be forecast then only a genuine risk premium or a relationship true by construction can pay. We tested both. The dollar index is an exact deterministic function of its six constituents and the index future shares an expiry code with the five currency futures on 98.2 per cent of sessions, so a matched synthetic basket is a no arbitrage relationship. It is also efficiently priced: the convergence trade returns a Sharpe ratio of $+0.09$ gross and -0.20 at one basis point per leg, because six legs are required against a spread whose daily change has a standard deviation of five basis points.

Carry is the more serious candidate. The roll gap between the expiring and deferred contract is the market's own quoted interest differential, which we validated against Federal Reserve interbank differentials to within 0.43 percentage points per currency. On the fifteen year futures panel it returns a Sharpe ratio of 0.377 and the mechanism verifies: of the 1.34 per cent annual return, 1.10 per cent is the differential actually held and 0.23 per cent is spot movement, which is the forward premium puzzle behaving as described. Table 10 shows why we do not claim it.

Table 10. Term structure carry on two samples. The futures panel begins in December 2010, immediately after the 2008 carry drawdown. Extending the sample to April 2002 using Federal Reserve spot and OECD interbank rates for all six constituents reverses the sign.

Sample	Currencies	Sharpe
Futures panel, 2011 to 2025	5	+0.377
Federal Reserve spot, 2002 to 2026	6	−0.249
<i>Sub periods of the long sample</i>		
2002 to 2007	6	+0.418
2008 to 2013	6	−0.760
2014 to 2019	6	−0.246
2020 to 2026	6	−0.022

The canonical factor set on the long sample gives carry -0.249 , momentum $+0.028$ and value $+0.096$, with an equal risk combination of -0.085 , against a noise ceiling whose ninetyeth percentile is $+0.41$. Volatility targeting, the one quantity that is forecastable, changes the Sharpe ratio by -0.009 on average across seven series over 27 years and improves three of them, which is inside the bootstrap null. What it does deliver is a reduction in the volatility of realised volatility from 0.025 to 0.015 . That is risk control and not return generation.

We report no profitable strategy. The reason is structural rather than a failure of effort. There is no directional predictability, the only candidate risk premium in this universe is negative over the full available sample and five to six highly correlated currencies with a common factor explaining 54.8 per cent of variance cannot support a cross sectional strategy.

7. A Reporting Protocol

Three additions, none expensive.

Screen the inputs. Report the binomial statistic of Equation (1), the frequency with which the open equals the previous close and the count of sessions violating the open, high, low, close ordering. These cost one pass over the data and would have stopped the first failure before any strategy was run.

Declare the whole search. Report every parameter that was varied and the total count of configurations evaluated, including those discarded. Report the ensemble across each family alongside the selected configuration, since an ensemble selects nothing and needs no correction.

Report the noise ceiling. Run the complete search once on block bootstrapped returns and report the best Sharpe ratio it produces. Any result below that number is indistinguishable from luck at that search size. In this study the ceiling was 0.71 to 0.75 for the 1,312 configuration search, 0.38 for a five strategy carry family and 0.41 for the canonical factor set. Every real result fell below its own ceiling.

8. Limitations

The evidence is one currency complex and two vendors. Nothing here generalises to free data as a category or to this vendor across other instruments. The mechanism behind the defect is inferred rather than observed: a stale print, a first electronic tick and a misfielded session extreme are all consistent with what we see and we cannot inspect the vendor's pipeline. The commercial series is itself a redistributor rather than the exchange and its closing field is a last price whose settlement status we could not confirm. The screen is validated against a single instrument. The noise ceiling is a presentation of an existing idea rather than a new statistic. Finally, the absence of a profitable strategy is a statement about six daily G10 series over fifteen to twenty seven years and not about foreign exchange markets in general; emerging market carry, options and intraday data all lie outside what we can test.

9. Conclusions

Two failures sit outside the reach of the standard defence of a backtest. A corrupted input is not merely invisible to the overfitting diagnostics, it is actively certified by them, because a stationary data defect persists out of sample exactly as a real anomaly does. An undeclared search space causes the correction to be applied to the wrong denominator and a strategy reporting $t = 3.09$ can fail every test once the full search is stated. Both failures are cheap to detect. A one line binomial test on the location of the open catches the first and a single run of the complete search on block bootstrapped returns catches the second, because it tells the analyst what their own search produces on data that contains nothing.

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